# Note 3 Stationary Processes

## 1 Describe a Time Series

Let $\{X_t\}_{t=1}^n$ be a time series of length $n$, if we want a complete description about $\{X_t\}$, we have to know the joint probability distribution of $(X_1, X_2, \cdots, X_n)$.
However, the joint distribution is difficult to evaluate most of the time. We instead use the first two moments to describe $\{X_t\}$.

- Mean:
$$\mu_t = \mathbb{E}(X_t) = \int_{\infty}^{\infty} x f_t(x) dx,$$
The mean describes the tendency of the central location of a time series.

- Autocovariance functions (ACVF):
$$\gamma(t, s) = \text{Cov}(X_t, X_s) = \mathbb{E}[(X_t - \mu_t)(X_s - \mu_s)].$$
The autocovariance functions describe the linear dependence structure of a time series.

## 2 Stationarity

### 2.1 Strictly Stationary Process

A series $\{X_t\}$ is called **strictly stationary** if the random vectors $(X_1, X_2, \cdots, X_n)$ and $(X_{h+1}, X_{h+2}, \cdots, X_{h+n})$ have the same probability distribution for all $n \geq 1$ and all integer $h$.

Note: Intuitively, strict stationarity means shifting $h$ does not alter the statistical properties of a time series.
If a time series consists of uncorrelated random variables with mean 0 and a constant variance, $\sigma_W^2$, we say it is **white noise**, usually denoted as $\{W_t\}$.
Moreover, if random variables are normally distributed, we call it **Gaussian white noise**, usually denoted as $\{Z_t\}$. Gaussian white noise will be intensively used later this semester.

## 2.2 Weakly Stationary Process

In practice, strictly stationarity is usually difficult to check. A weaker form of stationarity is the weak stationarity.

A series $\{X_t\}$ is called **weakly stationary** if

1) $\mu_t = \mathbb{E}(X_t)$ does not depend of time $t$;

2) $\gamma(t, s) = \text{Cov}(X_t, X_s)$ only depends on $t - s$.

- In this course, when we say "stationary", we mean weakly stationary.
- Strictly stationary $\Rightarrow$ weakly stationary, but not vice versa.
- For notation simplicity,

The **lag-$h$ autocorrelation** function of a stationary time series is

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)}$$

Some properties of autocovariance/autocorrelation functions of a stationary time series:

- Autocovariance functions are symmetric about zero, that is $\gamma(h) = \gamma(-h)$.

- $-1 \leq \rho(h) \leq 1$.

## 2.3 Some Examples

**Example 2.1.** Verify a white noise $\{W_t\}$ is stationary.

**Example 2.2.** Let $\{W_t\}$ be a white noise with mean 0 and constant variance $\sigma_W^2$, and

$$Y_t = \frac{W_t + W_{t-1}}{2}.$$

Verify $\{Y_t\}$ is stationary.

**Example 2.3.** A **random walk** $\{X_t\}$ is given by

$$X_t = X_{t-1} + W_t,$$

where $\{W_t\}$ is white noise. Is random walk stationary?

# 3 Estimations

Although the theoretical mean and autocovariance functions are useful for described properties of hypothesized models, most of the analyses must be performed using sampled data.

## 3.1 Estimation of $\mu$

If a time series $\{X_t\}$ is stationary, the mean $\mu$ can be estimated by the sample mean

$$\bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}.$$

- $\bar{X}$ is an unbiased estimator of $\mu$.
- $\mathrm{Var}(\bar{X}) =$

## 3.2 Estimation of $\gamma(\cdot)$ and $\rho(\cdot)$

The autocovariance function, $\gamma(h)$, is estimated by the **sample autocovariance function**, $\hat{\gamma}(h)$,

$$\hat{\gamma}(h) = \frac{\sum_{t=1}^{n-h}(x_{t+h} - \bar{x})(x_t - \bar{x})}{n}.$$

- The sample autocovariance function is also symmetric, that is $\hat{\gamma}(h) = \hat{\gamma}(-h)$.
- $\hat{\gamma}(h)$ is a **biased** estimator of $\gamma(h)$, but is asymptotically unbiased. That is,

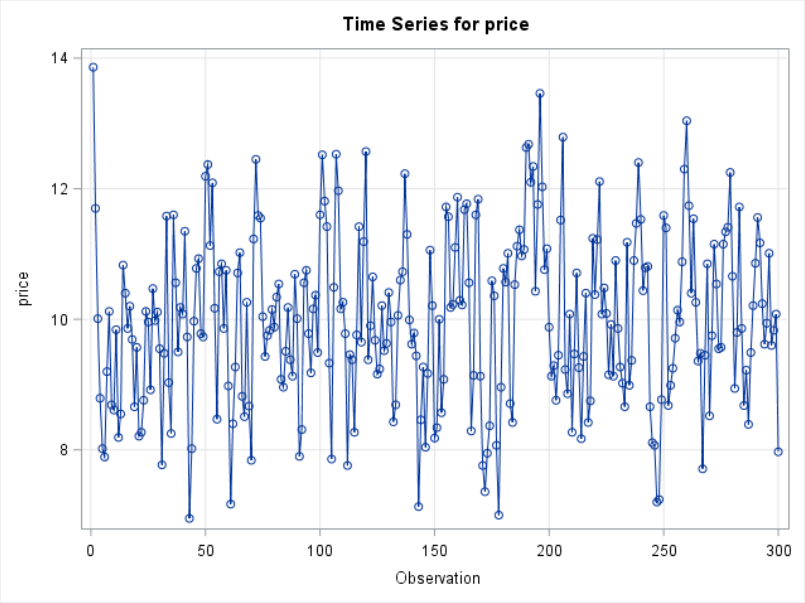
$$\mathbb{E}[\hat{\gamma}(h)] \neq \gamma(h), \quad \text{but } \lim_{n\to\infty} \mathbb{E}[\hat{\gamma}(h)] = \gamma(h)$$

The autocorrelation function, $\rho(h)$, is estimated by the sample autocorrelation function, $\hat{\rho}(h)$,

$$\hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}.$$

- By Bartlett's formula, $\text{Var}(\hat{\rho}(h)) \approx$

**Example 3.1.** Below is the daily price of a stock in 300 consecutive days. Assume this series is stationary.




(a) Find its autocorrelation functions.

(b) Is $\rho(1) = 0$? What about $\rho(2)$?

(c) Estimate the mean of the series.

(d) Is $\mu = 10$?

(e) Suggest a model for the series?

# 4 Check Stationarity

We have introduced stationary process. But in practice, how can we determine if a series is stationary or not?

1. Plot.

   A stationary series does not have any upward or downward trend or seasonal effects. Mean or variance are consistent over time.

   A non-stationary series shows trends, seasonal effects, and other structures depend on time.

2. Statistical tests: Augmented Dickey-Fuller test is a common choice. The detail of this test is skipped for now since it involves some topics we haven't covered yet.

   $H_0$ : the time series has a unit root, meaning it is non-stationary.

   $H_1$ : the time series does not have a unit root, meaning it is stationary.

**Example 4.1.** Perform augmented Dickey-Fuller test for the stock price series.